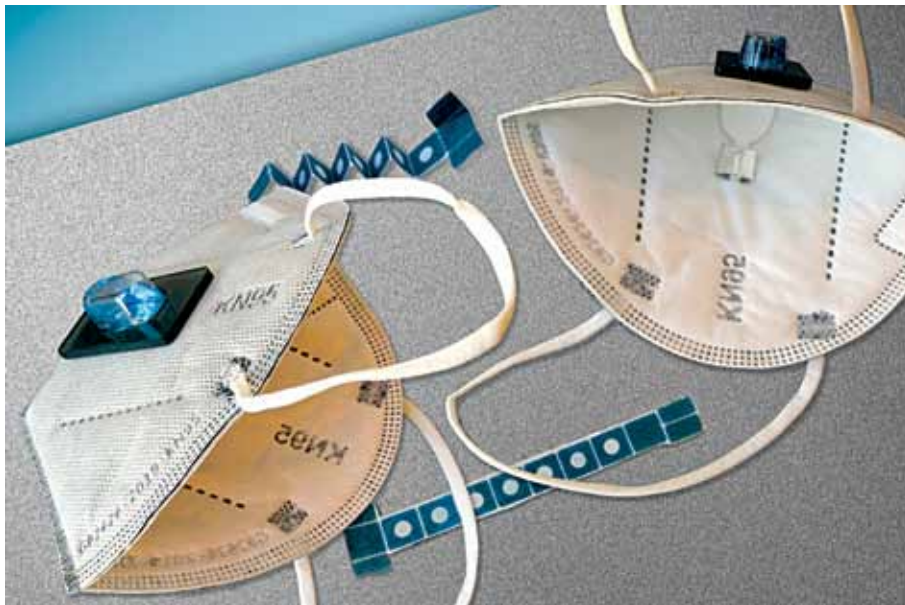


New face mask detects Covid-19 infection



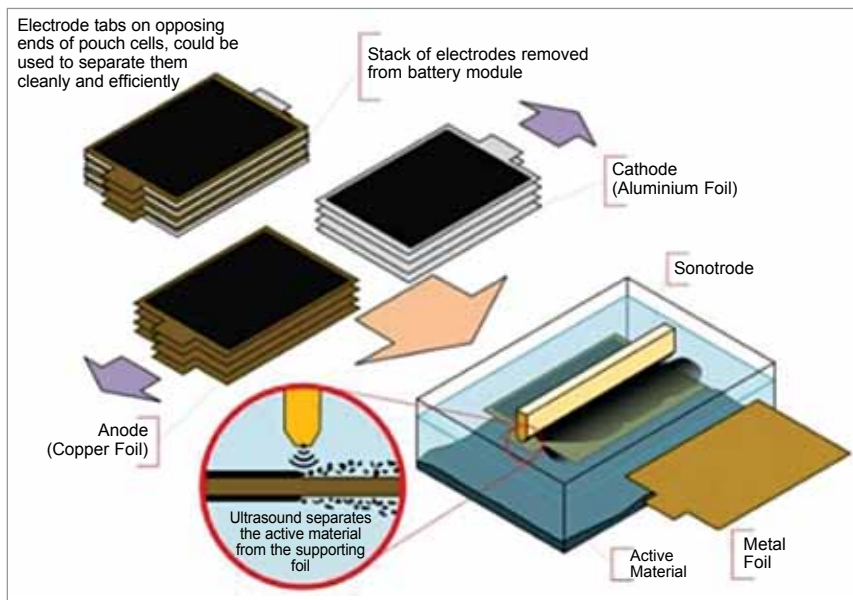
A prototype face mask for diagnosing Covid-19 within about 90 minutes (Credit: <https://news.mit.edu>)

Engineers at MIT and Harvard University have designed a novel face mask that can diagnose the wearer with Covid-19 within about 90 minutes. The masks are embedded with tiny, disposable sensors that can be fitted into other face masks and could also be adapted to detect other viruses. The sensors could be incorporated into not only face masks but also clothing such as lab coats. The sensors are based on freeze-dried cells that can remain stable for many months. When activated by water, they interact with their target molecule, which can be any RNA or DNA sequence, as well as other types of molecules, and produce a signal such as a change in colour.

Groundbreaking approach for recycling batteries

Current recycling methods for lithium-ion batteries typically feed end-of-life batteries into a shredder or high-temperature reactor, which are energy intensive and inefficient. Researchers at the University of Leicester have now developed a new method to recycle electric vehicle batteries using a ground-breaking new approach involving ultrasonic waves to separate out valuable materials from electrodes. So, now the materials such as lithium, nickel, manganese, and cobalt can be fully recovered from batteries at the end of their life.

Diagram showing the ultrasonic process which assists in the delamination of lithium-ion batteries (Credit: <https://le.ac.uk>)



Intelligent robotic arm that can touch like humans

Researchers at the National University Singapore have developed a smart foam material that has the ability to heal and repair itself. So, engineers will now be able to develop robots

and smart prosthetic devices that feel more human-like. To replicate the human sense of touch, the researchers infused the foam material with microscopic metal particles and added